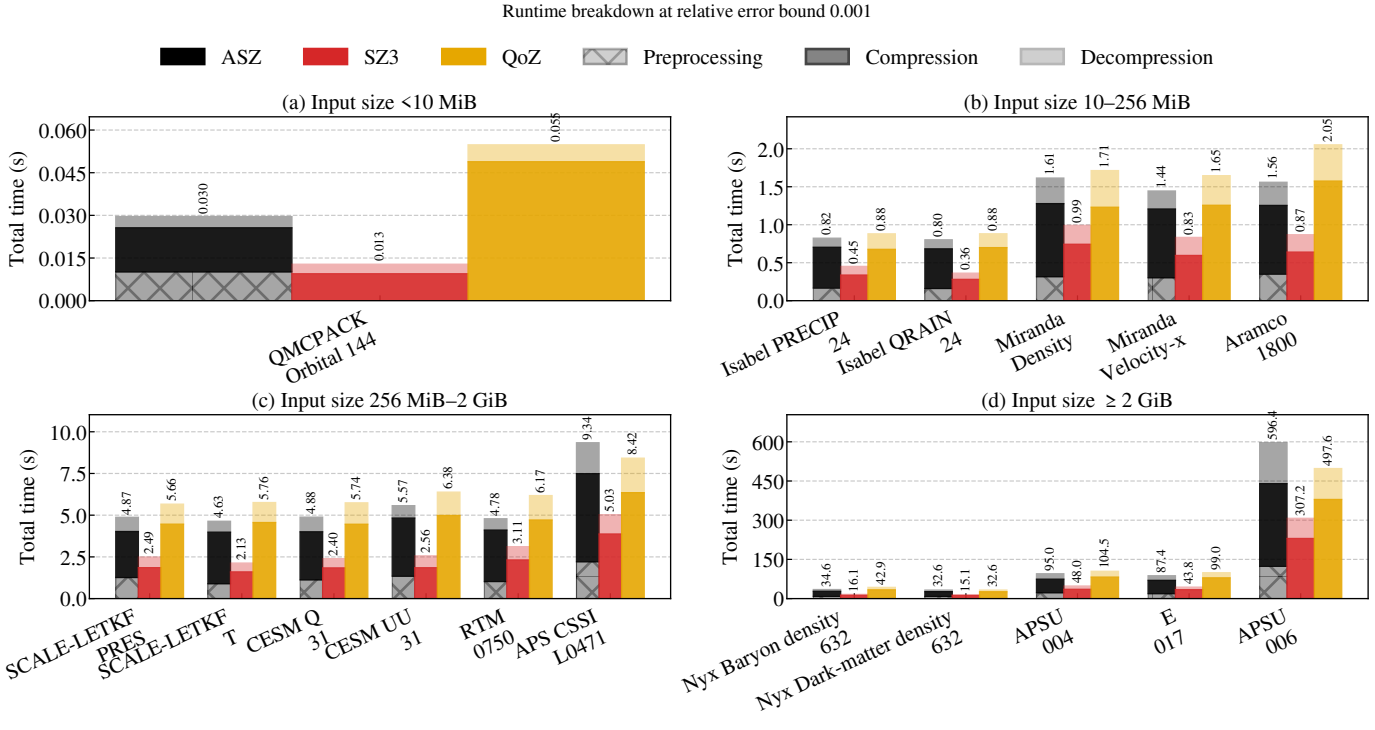
Runtime breakdown at relative error bound 0.001
ASZ
SZ3
QoZ
Preprocessing
Compression
Decompression
(a) Input size <10 MiB
Total time (s)
0.060
0.045
0.030
0.015
0.000
0.030
0.013
0.055
QMCPACK
Orbital 144
(b) Input size 10–256 MiB
Total time (s)
2.0
1.5
1.0
0.5
0.0
0.82
0.45
0.88
0.80
0.36
0.88
1.61
0.99
1.71
1.44
0.83
1.65
1.56
0.87
2.05
Isabel PRECIP
24
Isabel QRAIN
24
Miranda
Density
Miranda
Velocity-x
Aramco
1800
(c) Input size 256 MiB–2 GiB
Total time (s)
10.0
7.5
5.0
2.5
0.0
4.87
2.49
5.66
4.63
2.13
5.76
4.88
2.40
5.74
5.57
2.56
6.38
4.78
3.11
6.17
9.34
5.03
8.42
SCALE-LETKF
PRES
SCALE-LETKF
T
CESM Q
31
CESM UU
31
RTM
0750
APS CSSI
L0471
(d) Input size ≥ 2 GiB
Total time (s)
600
450
300
150
0
34.6
16.1
42.9
32.6
15.1
32.6
95.0
48.0
104.5
87.4
43.8
99.0
596.4
307.2
497.6
Nyx Baryon density
632
Nyx Dark-matter density
632
APSU
004
E
017
APSU
006